

DATA100 Midterm Version 1 (pink)

Tuesday, October 6, 2026

Instructions: Put your name, WLU id, **and section** on the answer sheet (which is separate).

Answer all questions on the answer sheet.

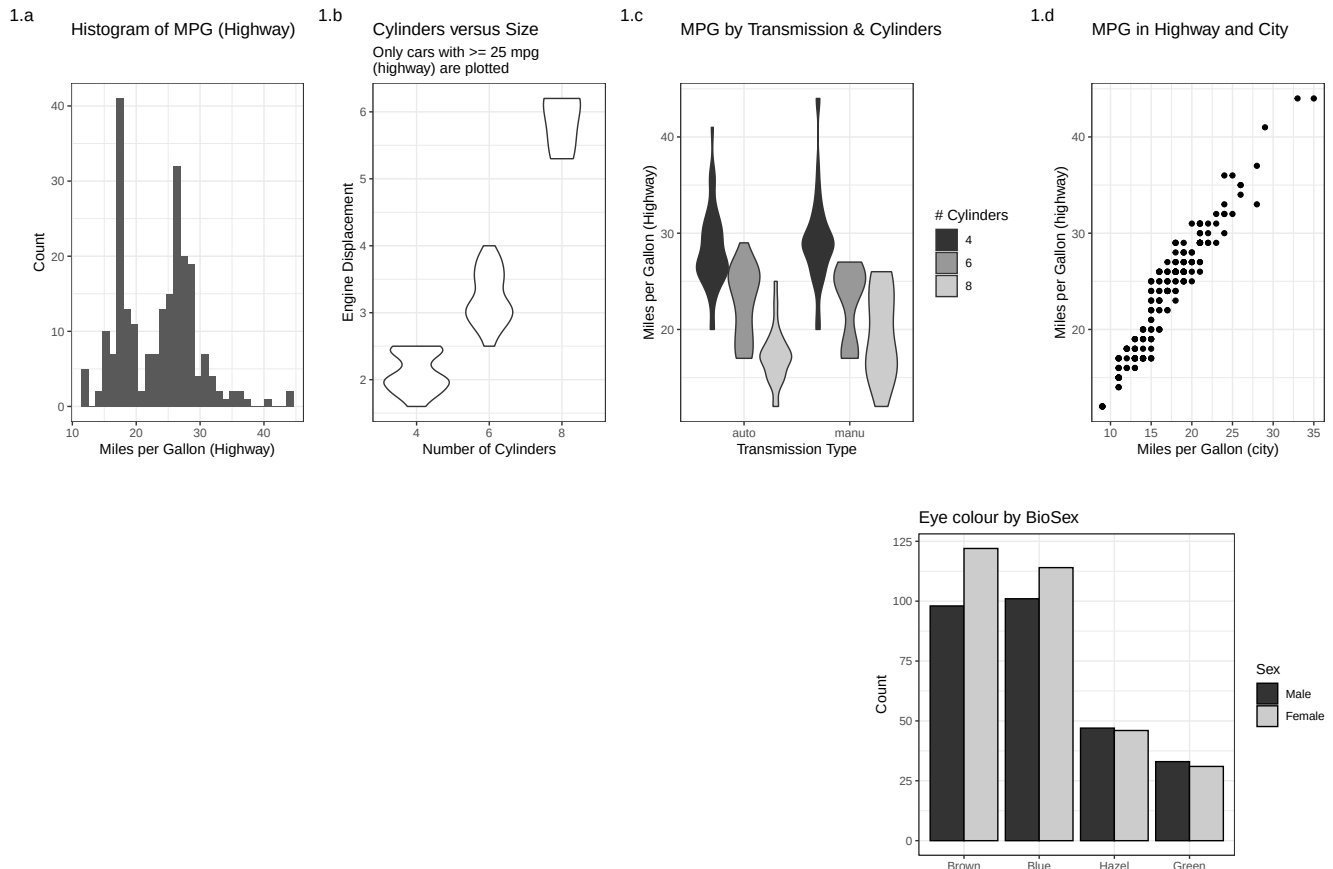
Only the **answer sheet** will be graded. No answers on this booklet will be considered. You may use this booklet for scrap paper.

There are 8 pages, with questions printed on the front and back of each sheet.

Multiple Choice Questions

Each question below is worth one mark. Answer on the **answer sheet** provided.

1. Which of the following plots provides information about which kinds of cars have the highest miles per gallon (highway)?

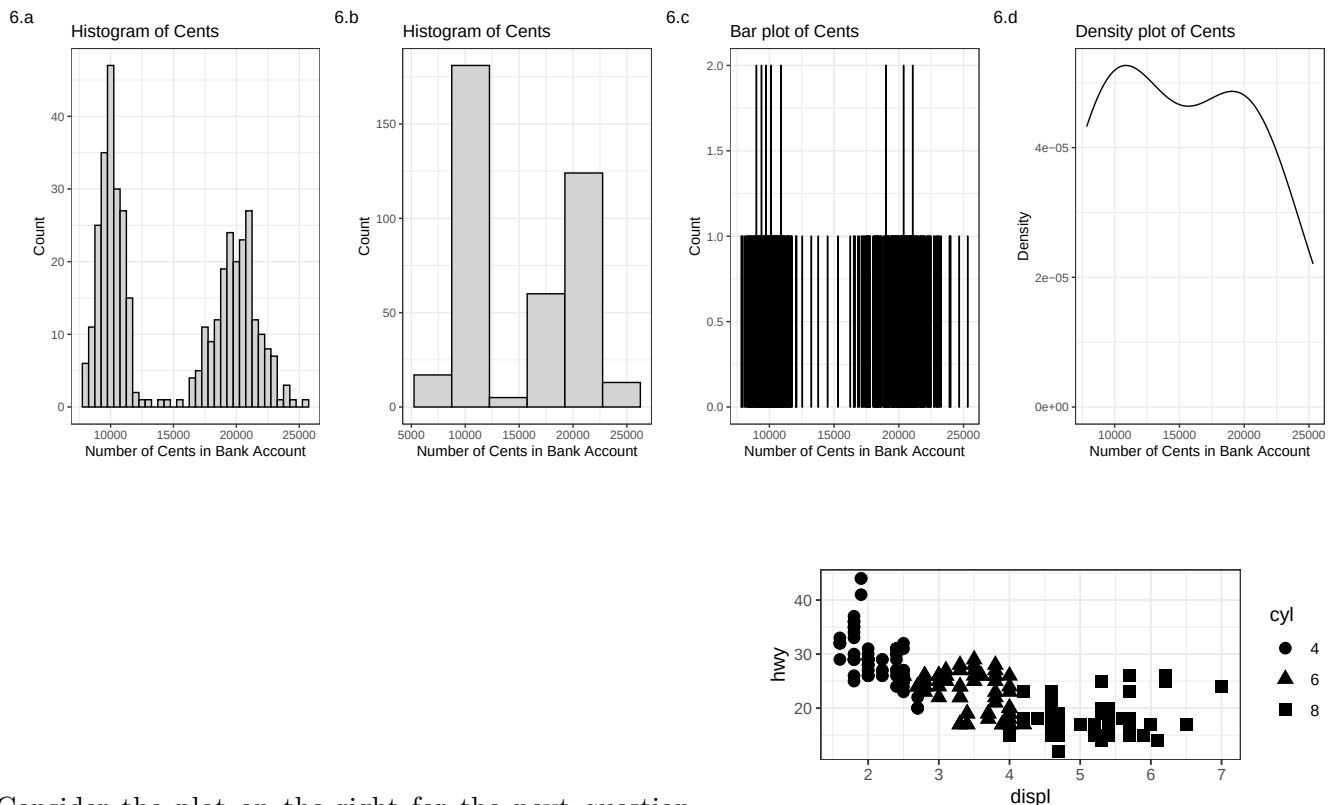


2. Which of the following is **NOT** a valid interpretation of the plot on the right?
- There is an overall decreasing trend when we move from brown to green eyes.
 - There were more female subjects than male subjects
 - There were more female subjects with blue and brown eyes than male subjects
 - Brown and blue eyes appear to be overall more common than hazel and green
4. In `ggplot2`, what is the best description of a `geom`?
- Declaring the way a data set will be used, e.g. by assigning columns to a dataframe.
 - The way data are assigned to elements of a plot, e.g. assigning `mpg` to the y axis or assigning `cyl` to the colour scheme.
 - The way elements of a plot are actually drawn, e.g. by declaring that a violin plot will be used.
 - The way additional plot elements, such as arrows or text, are added to the plot.

3. Consider the data on the right. Which of the following plots would provide meaningful information?

id	name	height	year	grade
1	Bonk	1.2	1	56
2	Lilah	0.4	NA	NA
3	June	1.4	3	88
4	Judy	1.6	2	75
5	Julie	1.7	1	72
6	Jodie	1.9	1	83
7	Jill	1.4	2	81

5. Consider a data set which records the number of cents in each person's bank account, where each bank account can only have a whole number of cents (no decimals). What type of variable is this?
- Qualitative - Categorical
 - Qualitative - Ordinal
 - Quantitative - Continuous
 - Quantitative - Discrete
6. Consider the data from Question 5; the number of cents in a person's bank account. Which of the following visualization provides the most useful information?



Consider the plot on the right for the next question. Note that the name of the data set is `mpg`, with columns `class` for the type of car (sedan, SUV, truck, etc.), `displ` for the size of the engine (displacement) in cubic feet, `hwy` for highway fuel efficiency in miles per gallon, and `cyl` for number of cylinders in the engine.

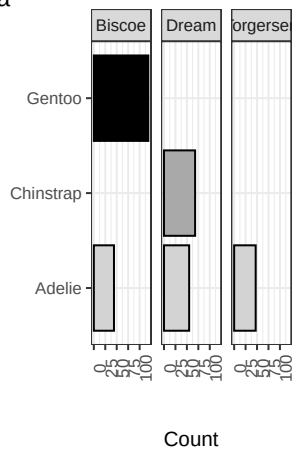
7. Which code produced the figure above?

- a. `ggplot(mpg) + aes(x = displ, y = hwy, shape = cyl) + geom_point()`
- b. `ggplot(mpg) + aes(x = hwy, y = displ, colour = cyl) + geom_point()`
- c. `ggplot(mpg) + aes(x = displ, y = hwy, colour = cyl) + geom_scatterplot()`
- d. `ggplot(mpg) + geom_scatterplot(x = hwy, y = displ, shape = cyl)`

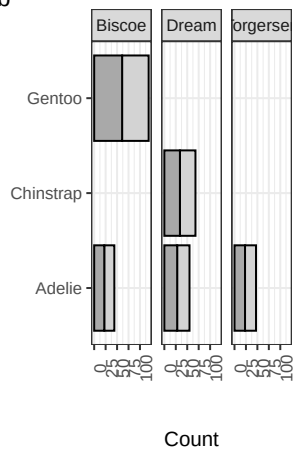
8. Consider the following code. Which figure would it produce?

```
ggplot(penguins) +
  aes(x = species, fill = species) +
  geom_bar() +
  facet_wrap(~ island) +
  coord_flip() +
  labs(
    x = NULL, y = "Count"
  )
```

8.a



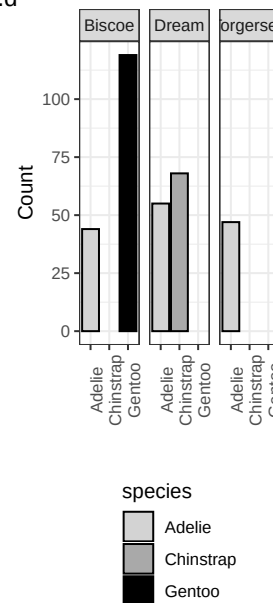
8.b



8.c



8.d



Short Answer Questions

16. Plot the following data on the axes provided in the answer sheet. Choose whichever plot you believe is most appropriate.

```
# A tibble: 1 x 3
  grade class student
<dbl> <chr>   <chr>
1     1 chemistry me
```